ModelIt!

Student Activity Pack

How Fast Does Ice Melt?

Exploring Reversible Changes in Matter

Grade 2 | NGSS 2-PS1-4 | Cause and Effect

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What happens to an ice cube when you take it out of the freezer?

2 Can you turn water back into ice? How?

3 Draw what you think happens to an ice cube when it melts — where does the ice go?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Surrounding Temperature | Melting Speed | Ice Size

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Warm Room

Surrounding Temperature set to warm

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Cold Spot

Surrounding Temperature set to cold

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Hot Sunshine

Surrounding Temperature set to very hot

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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> Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Insulation — Materials wrapped around the ice that slow heat transfer — like a blanket for the ice cube

Ice Surface Area — How much of the ice is touching warm air — crushed ice melts faster than a big cube because more surface touches the warm air

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

What makes ice melt fast or slow — and can you turn the water back into ice again? Use evidence from your simulation to support your answers.

1 Why does ice melt faster in some places than others?

2 What makes melting ice different from baking a cookie — can you undo both changes?

8 STEM Challenge: Design an Ice Keeper

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YOUR ENGINEERING MISSION

Build a container that keeps an ice cube frozen as long as possible — the team whose ice lasts longest wins!

THE SCENARIO

A lunch company wants to keep food cold during delivery without using electricity. Your team has been challenged to design an Ice Keeper — a container that slows down melting as much as possible. Use what you learned about temperature and melting to make your design.

Engineering Design Thinking

1. WHAT INSULATING MATERIALS WILL YOU TEST — FABRIC, COTTON, BUBBLE WRAP, FOIL, OR NEWSPAPER?

2. SHOULD YOU USE ONE LAYER OR MANY LAYERS AROUND THE ICE?

3. HOW WILL YOU COMPARE YOUR ICE KEEPER TO JUST LEAVING THE ICE IN THE OPEN?

4. HOW LONG DO YOU PREDICT YOUR ICE WILL LAST COMPARED TO THE OPEN ICE CUBE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Refrigeration Engineers design cooling systems for homes, restaurants, hospitals, and trucks that deliver food. They keep things cold so food stays fresh and medicine stays safe. They earn $50,000–$90,000/year.